

Ordered, Disordered and Hexatic Phases of Graphene

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1 Introduction

Two-dimensional (2D) crystalline materials exhibit an important sensitivity to structural imperfections. In 2D systems, topological defects such as dislocations and grain boundaries play a central role in determining both structural and functional properties [1], as their elastic fields extend over large distances. Graphene is a particularly relevant physical realization of this paradigm. Large-area graphene samples are commonly polycrystalline, consisting of multiple crystallites with different orientations and separated by grain boundaries. These grain boundaries are not only geometric imperfections but constitute extended topological defects composed of periodic arrays of dislocations. They have been extensively investigated, revealing a strong dependence on the relative misorientation between neighboring grains [2].

From a theoretical perspective, grain boundaries in graphene can be understood as arrays of dislocations whose spacing is controlled by the misorientation angle. In the low-angle regime, this description naturally leads to the Read–Shockley form of the grain-boundary energy [3]. Grain boundaries also raise important questions concerning the statistical mechanics of orientational degrees of freedom in polycrystalline graphene. In particular, the possibility of nontrivial orientational phases. In two dimensions, such phases are known to be strongly constrained by topology, as conventional long-range order is prohibited in the presence of continuous symmetries [4]. Instead, orientational order may persist only in a quasi-long-range form.

Polycrystalline graphene is closely related to the class of two-dimensional systems described by the Kosterlitz–Thouless–Halperin–Nelson–Young (KTHNY) theory of melting [5–8]. While KTHNY theory was originally developed for atomic and molecular crystals, its underlying principles are more general and can, in principle, apply to coarse-grained orientational degrees of freedom.

In this work, we explore these ideas through a coarse-grained model of polycrystalline graphene. Each crystallite is represented by a single orientational variable, and the energetic cost of misorientation between neighboring grains is encoded using a Read–Shockley interaction. An additional substrate coupling term favors global alignment [9]. Using Monte Carlo simulations, we investigate the thermal behavior of this model and characterize the resulting orientational phases through bond-orientational correlation functions and susceptibilities. We first reproduce well-established results for the two-dimensional XY model [5] and then analyze the polycrystalline model, demonstrating the emergence of a phase transition. The present study thus verify that the KTHNY theory can be applied to this coarse-grained model of polycrystalline graphene.

2 Polycrystalline Model

2.1 Hamiltonian

The project focuses on a two-dimensional system of grains (crystallites) arranged on a hexagonal lattice structure. Each grain has an orientation angle, denoted by θ_i , which represents its deviation from a fixed reference orientation. This configuration corresponds to a graphene sheet placed on a hexagonal substrate.

In graphene, grain boundaries are known to consist of periodic arrays of dislocations whose elastic interactions determine the energetic cost of misorientation between neighboring crystallites [2]. The distribution of grain orientations is governed by two competing interactions: the substrate coupling [9], which tends to align the grains with the substrate, and the grain boundary energy, which favors local alignment between neighboring grains. This interaction is expressed according to Read-Shockley model [3]. The total Hamiltonian $\mathcal{H}$ of the system can be written as:

$$\mathcal{H} = \frac{\epsilon}{\rho} \sum_i \theta_i^2 + \frac{\gamma}{3\rho} \sum_i \sum_{j \in nn_i} |\Delta\theta_{ij}| (A - \log |\Delta\theta_{ij}|) \quad (1)$$

Where ϵ is the substrate coupling strength, γ is the grain boundary energy coefficient, ρ is the grain density and A is a constant related to the material properties. The term $\Delta\theta_{ij} = \theta_i - \theta_j$ represents the difference in orientation angles between neighboring grains i and j , with the summation over nn_i indicating that we only consider nearest neighbors. From now, ρ and A are set to 1 for simplicity.

2.2 Observables

We define several observables to characterize the phases in the system as described in the KTHNY theory [5–8]. To characterize the orientational order of the system, the local bond-orientational order parameter $\psi_6(\mathbf{r}_i)$ for each grain i is defined as:

$$\psi_6(\mathbf{r}_i) = \frac{1}{n_i} \sum_{j \in nn_i} e^{i6\theta_{ij}} \quad (2)$$

Where n_i is the number of nearest neighbors of grain i , and θ_{ij} is the angle between a reference axis and the bond connecting grain i to its neighbor j . This parameter captures the six-fold rotational symmetry characteristic of the hexagonal lattice structure. Here, a coarse-grained approximation is used. The local bond orientation of each grain is fully determined by its orientation angle θ_i , so the local bond-orientational order parameter can be simplified to:

$$\psi_6(\mathbf{r}_i) = e^{i6\theta_i} \quad (3)$$

The bond-orientational correlation function $G_6(r)$ between grains separated by a distance r is defined as:

$$G_6(r) = \langle \psi_6(\mathbf{r}_i) \psi_6^*(\mathbf{r}_j) \rangle_{|\mathbf{r}_i - \mathbf{r}_j| = r} \quad (4)$$

Where the average is taken over all pairs of grains (i, j) separated by distance r . The behavior of

$G_6(r)$ provides information about the orientational order of the system. In a hexatic phase, $G_6(r)$ is expected to decay as a power law:

$$G_6(r) \sim r^{-\eta_6} \quad (5)$$

Where η_6 is a critical exponent characterizing the decay of orientational correlations. At the transition between hexatic and disordered phases, η_6 reaches a universal value of $1/4$. In contrast, in a disordered phase, $G_6(r)$ decays exponentially:

$$G_6(r) \sim e^{-r/\xi_6} \quad (6)$$

Where ξ_6 is the orientational correlation length. In an ordered phase, $G_6(r)$ must be equal to a constant for large r , indicating long-range orientational order. In the limiting case $\gamma = 0$, grain orientations become independent, leading to an exact expression for $G_6(r)$ [10]:

$$G_6(r) = \exp\left(-18\rho\frac{k_B T}{\epsilon}\right) \quad (7)$$

Finally, the susceptibility associated with the bond-orientational order parameter, χ_6 , is defined as [11]:

$$\chi_6 = N\beta \left(\langle |\Psi_6|^2 \rangle - \langle |\Psi_6| \rangle^2 \right) \quad (8)$$

Where N is the total number of grains in the system, $\beta = \frac{1}{k_B T}$ and $\Psi_6 = \frac{1}{N} \sum_i \psi_6(\mathbf{r}_i)$ is the global bond-orientational order parameter. This susceptibility measures the fluctuations in the bond-orientational order parameter. At the hexatic to disordered phase transition, χ_6 is expected to diverge. In finite systems, this divergence manifests as a pronounced peak scaling with system size.

3 Simulation Method

To study the behavior of the polycrystalline graphene system described by the Hamiltonian above, a Monte Carlo simulation approach using the Metropolis algorithm is employed. The simulation is performed on a two-dimensional hexagonal lattice of size L with $N = L \times L$ sites, where each lattice site represents a grain with an associated orientation angle θ_i . The equilibrium configurations are sampled using the Metropolis algorithm and deterministic angular reflection moves inspired by over-relaxation techniques used in the XY model.

3.1 Local Metropolis Updates

During a Monte Carlo sweep, N update attempts are made. At each attempt, a random grain i is selected, and a new orientation angle is proposed according to:

$$\theta'_i = \theta_i + \delta\theta, \quad \delta\theta \in [-\Delta, \Delta] \quad (9)$$

Where Δ is the maximum angular step size. The proposed move is accepted with the standard Metropolis probability:

$$P_{\text{accept}} = \min\left(1, e^{-\beta\Delta E}\right) \quad (10)$$

Where $\Delta E = \mathcal{H}(\theta'_i) - \mathcal{H}(\theta_i)$ is the energy difference resulting from the Hamiltonian defined in Eq. (1). The parameter Δ is dynamically adjusted during the thermalization phase to maintain an acceptance ratio of approximately 50%.

3.2 Over-relaxation Updates

In addition to stochastic Metropolis moves, deterministic angular reflection updates are incorporated to enhance sampling efficiency. This method is inspired by over-relaxation techniques for the XY model [12]. In the XY model, such moves are microcanonical and preserve energy. However, due to the more complex interactions in our Hamiltonian, the energy is not strictly conserved. Therefore, the proposed over-relaxation move is accepted with the same Metropolis criterion as above. For a selected grain i , a local reference orientation is computed:

$$\mathbf{S}_i = \sum_{j \in nm_i} (\cos(6\theta_j); \sin(6\theta_j)) \quad (11)$$

$$\phi_i = \text{atan2}(S_{i,y}, S_{i,x}) \quad (12)$$

The proposed update is then:

$$\theta'_i = 2\frac{\phi_i}{6} - \theta_i \quad (13)$$

And the acceptance probability follows the Metropolis criterion. Although these moves do not constitute a microcanonical update, they help the system explore the configuration space more effectively.

3.3 Angular Domain and Boundary Conditions

Due to the sixfold symmetry of the graphene lattice, grain orientations are defined modulo $\pi/3$. Therefore, the orientation angles are restricted to the interval:

$$\theta_i \in \left[-\frac{\pi}{6}, \frac{\pi}{6}\right] \quad (14)$$

Which eliminates redundant configurations. Open boundary conditions are applied to the lattice, allowing grains at the edges to have fewer neighbors. This choice reflects the physical reality of finite graphene samples rather than an infinite periodic system.

3.4 Simulation Procedure

Each simulation is divided into a thermalization phase and a measurement phase. During the thermalization phase, the angular step size Δ is adjusted and the system is allowed to reach equilibrium. In the measurement phase, observables are recorded at regular intervals to ensure statistical independence.

Algorithm 1 Monte Carlo Simulation Procedure

```

1: Initialize  $\{\theta_i\} \in [-\frac{\pi}{6}, \frac{\pi}{6}]$  uniformly at random
2: for  $n_{steps\_therm}$  do
3:   Perform a standard Monte Carlo sweep
4:   Perform an over-relaxation sweep every  $n_{overrelax}$  steps
5:   Adjust  $\Delta$  to maintain an acceptance ratio of  $\sim 50\%$  every  $n_{tune}$  steps
6: end for
7: for  $n_{steps\_meas}$  do
8:   Perform a standard Monte Carlo sweep
9:   Measure and save observables every  $n_{measure}$  steps
10: end for

```

3.5 Measurements and Statistical Analysis

During the measurement phase, the bound-orientationnal correlation function $G_6(r)$, the global order parameter Ψ_6 , and the corresponding susceptibility χ_6 are computed. Measurements are separated by several Monte Carlo sweeps to reduce autocorrelation effects and ensure statistical independence.

4 Results

First weeks were dedicated to implement, optimize and test the Monte Carlo simulation code. Reaching an efficient and accurate code was essential before studying the polycrystalline model and it was a challenging task. Systems studied need a lot of Monte Carlo steps to reach equilibrium near transition. Optimize and parallelize the code was then very important and it took some time.

The first part of the results section focuses on validating the Monte Carlo simulation code by applying it to the well-known XY model. After that, the polycrystalline model is studied to know if the system can be described by the KTHNY theory.

4.1 Validation on the XY model

The first step was to validate the Monte Carlo simulation code by applying it to the classical XY model, which is a well-studied system. The Hamiltonian of the XY model is given by:

$$\mathcal{H}_{XY} = -J \sum_{\langle i,j \rangle} \cos(\theta_i - \theta_j) \quad (15)$$

Where J is the coupling constant, $J = 1$ for the rest of this part. The geometry used is the same hexagonal lattice as in the polycrystalline model. The thermodynamic behavior of the XY model is characterized by a Berezinskii-Kosterlitz-Thouless (BKT) [5] transition at a critical temperature T_{BKT} . Below this temperature, the system exhibits quasi-long-range order, while above it, the system is disordered. According to Sorokin [13], who studied different lattice geometries, the critical temperature for the XY model on a triangular lattice is approximately $T_{BKT} = 1.418$. The correlation function $G(r)$ was computed and the exponent $\eta(T)$ was extracted to find the transition temperature where $\eta(T_{BKT}) = 1/4$. Simulations were run for lattice sizes $L = 16, 32, 64$ and for temperatures ranging from $T = 1 \times 10^{-5}$ to $T = 2.0$. Correlation functions for $L = 64$ and their fits are shown in Fig. 1. The extracted exponent $\eta(T)$ as a function of temperature is shown in Fig. 2a. Finally, the susceptibility χ as a function of temperature for different lattice sizes is shown in Fig. 2b. Results are in good agreement with the expected behavior of the XY model. With the critical exponent condition $\eta(T_{BKT}) = 1/4$, the transition temperature is found to be $T_{BKT} = 1.42 \pm 0.03$, which is consistent with the value in the literature. Additionally, the susceptibility χ shows a peak just above the transition temperature, which becomes more pronounced with increasing lattice size, as expected. Under the critical temperature, the system exhibits quasi-long-range order, while above it, the system decays exponentially, confirming the disordered phase. At $T = 1 \times 10^{-5}$, the system is nearly perfectly ordered.

Under these observations, the Monte Carlo simulation code is validated for the XY model and can be used to study the polycrystalline model.

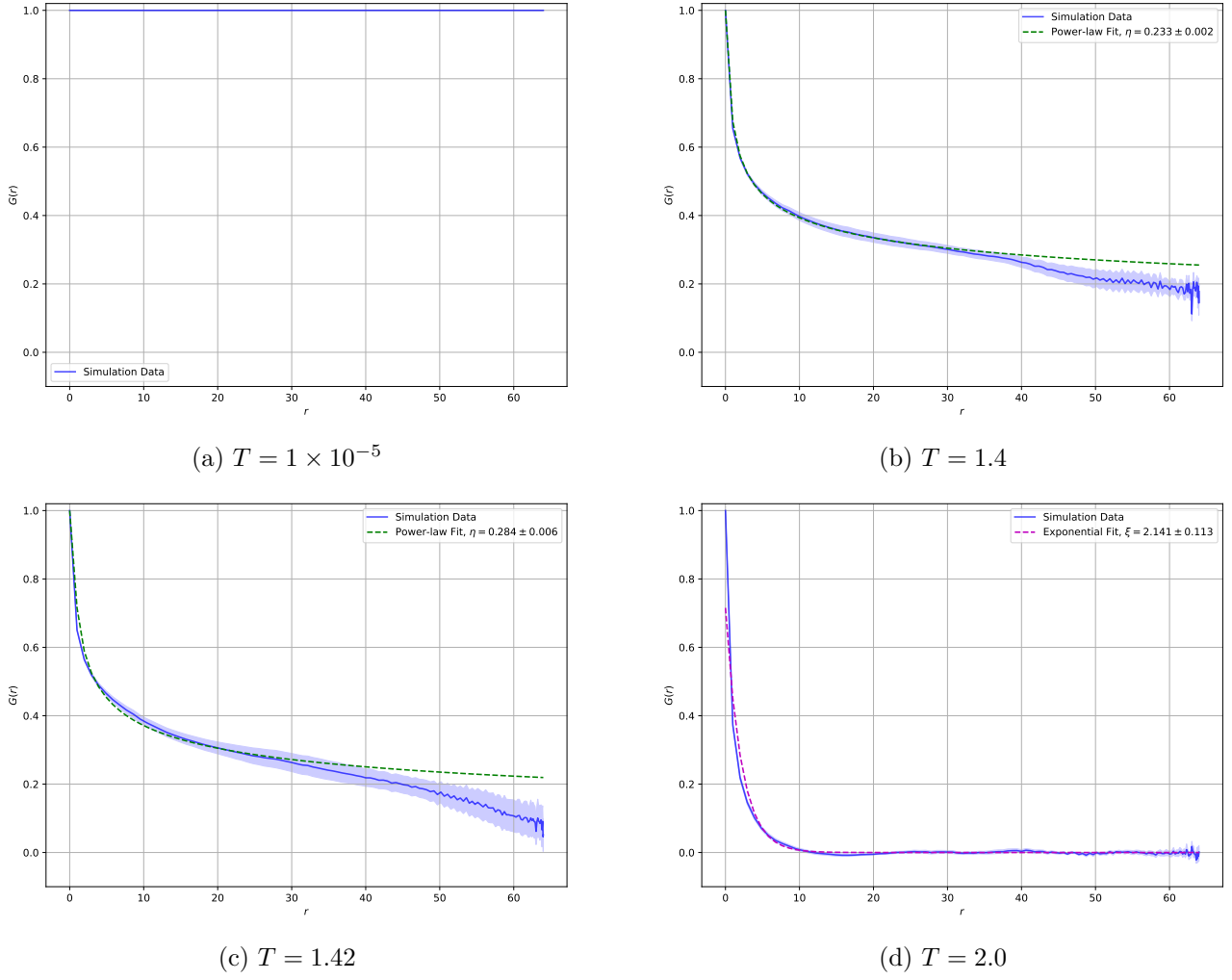


Figure 1: Correlation functions of the XY model at different temperatures with lattice size $L = 64$.

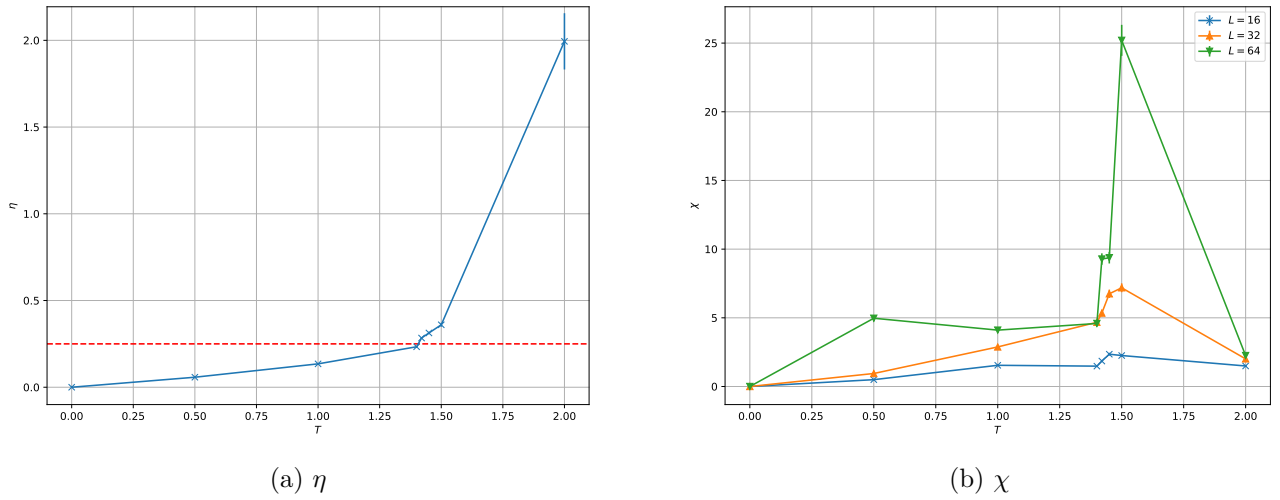


Figure 2: Exponent η and susceptibility χ of the XY model as a function of the temperature for $L = 64$, respectively, $L = 16, 32, 64$.

4.2 Polycrystalline Model without Grain Interactions

The first step was to verify the crystalline order of the system when there is no interaction between grains, i.e., $\gamma = 0$ and $\epsilon = 1$. Simulations were run for $L = 16, 32, 64$ and for temperatures ranging from $T = 1 \times 10^{-5}$ to $T = 1.0$. The measured correlation functions and their theoretical predictions are shown in Fig. 3 for $L = 64$. The susceptibility χ_6 as a function of the inverse lattice size L^{-1} is shown in Fig. 4. These results confirm that when there is no interaction between grains, the system behaves like a perfect crystal until it reaches a totally disordered state at high temperatures. Correlation functions match the theoretical predictions calculated with Eq. (7), and the susceptibility χ_6 remains constant with increasing lattice size, indicating long-range order or total disorder.

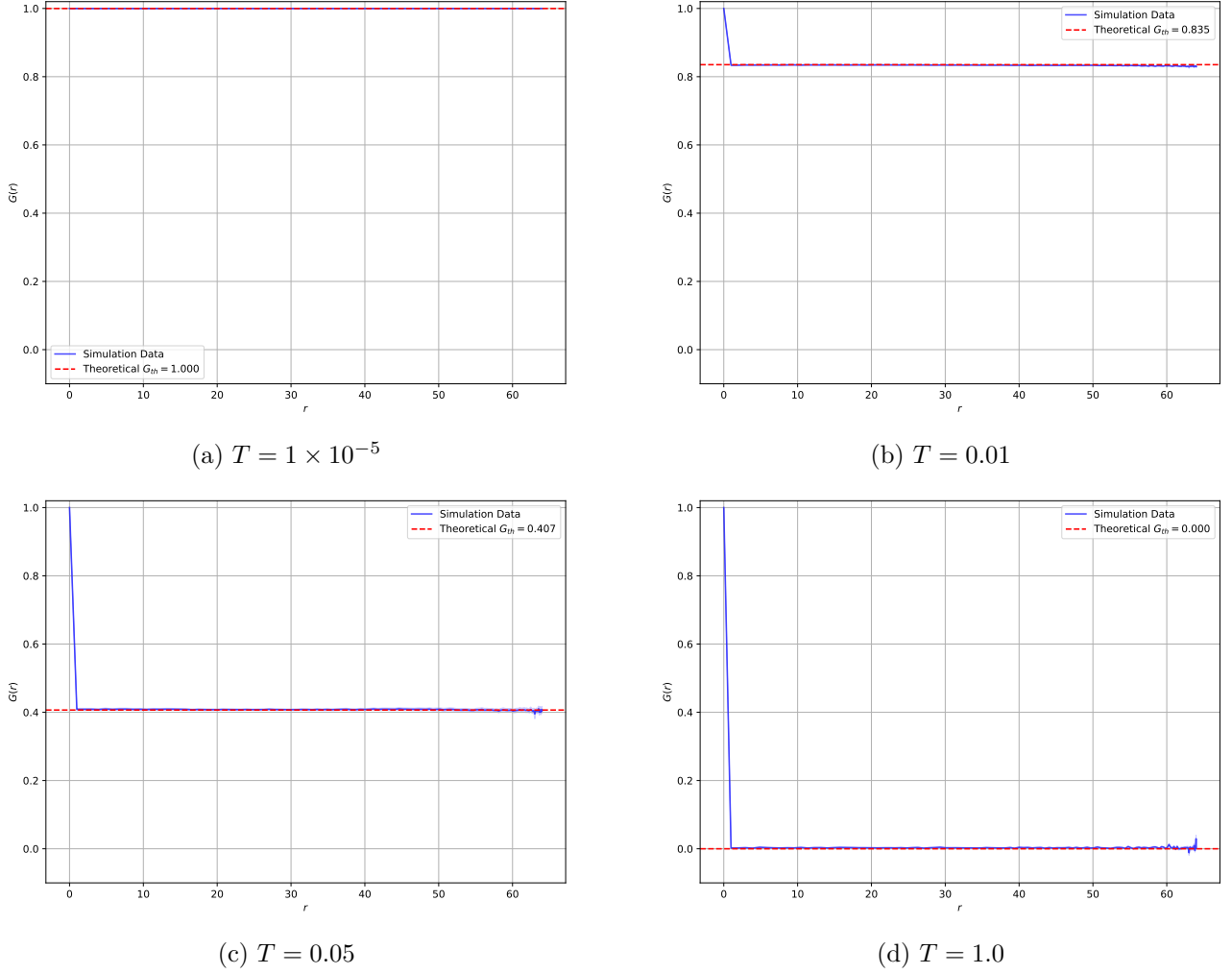


Figure 3: Correlation functions for the polycrystalline model with $\epsilon = 1.0$ and $\gamma = 0.0$ at different temperatures with lattice size $L = 64$.

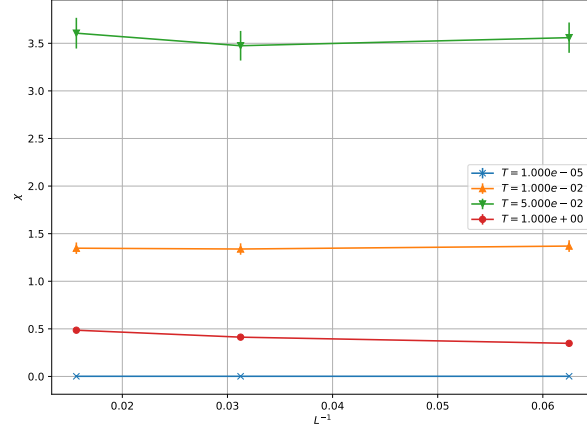


Figure 4: Susceptibility χ_6 as a function of the inverse lattice size L^{-1} for the polycrystalline model with $\epsilon = 1.0$ and $\gamma = 0.0$ at different temperatures.

4.3 Polycrystalline Model with Grain Interactions

The system was then studied with grain interactions, setting $\gamma = 1.0$ and $\epsilon = 0.0$. Simulations were run for $L = 16, 32, 64, 128, 256$ and for temperatures ranging from $T = 1 \times 10^{-5}$ to $T = 0.3$. The measured correlation functions for $L = 256$ and their fits are shown in Fig. 5 for different temperatures. To compare the configuration in different phases, final lattices of each temperature are shown in Fig. 8. The extracted exponent $\eta_6(T)$ as a function of temperature is shown in Fig. 6a. The susceptibility χ_6 as a function of temperature for different lattice sizes is shown in Fig. 6b. Finally, the susceptibility χ_6 as a function of the inverse lattice size L^{-1} at different temperatures near the transition and far from the transition is shown in Fig. 7. Results indicate that the system undergoes a phase transition consistent with a BKT transition. The correlation functions show a change from a power-law decay to an exponential decay as the temperature increases. The exponent η_6 reaches the critical value of $1/4$ at a transition temperature of $T = 0.195 \pm 0.005$. The susceptibility χ_6 exhibits a peak right above the transition temperature, which becomes more pronounced with increasing lattice size. At temperatures near the transition, the susceptibility increases exponentially with lattice size, while far from the transition, it remains relatively constant, indicating a transition to a disordered phase. The use of open boundary conditions introduce some finite-size effects, but the overall behavior is consistent with expectations for a BKT transition. To avoid these effects, larger lattice sizes are necessary but they require significantly more computational resources.

These findings support the applicability of the KTHNY theory to this polycrystalline model with grain interactions. However, the results are not very accurate yet, and further investigations with larger lattice sizes and more temperatures are needed to confirm these observations and refine the estimates of critical exponents and transition temperatures. Additionally, the existence of the hexatic phase between the ordered and disordered phases should be examined at higher lattice sizes with more thermalization steps to study attentively the behavior of the correlation functions at large distances.

The influence of the substrate coupling strength ϵ on the phase behavior should also be explored in future work to see how the competition between substrate coupling and grain interactions affects the phase transition. Parameters A and ρ in the Hamiltonian could also be varied to study their impact on the system's behavior.

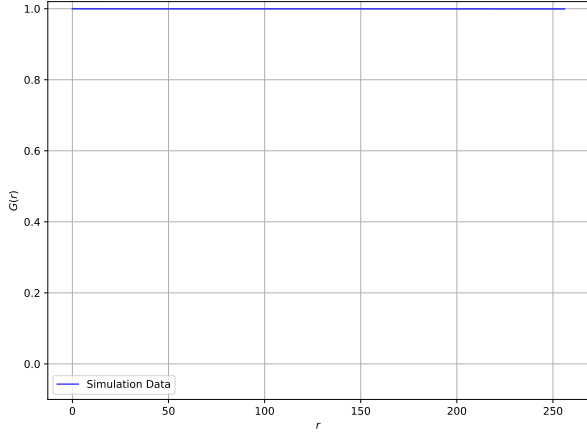
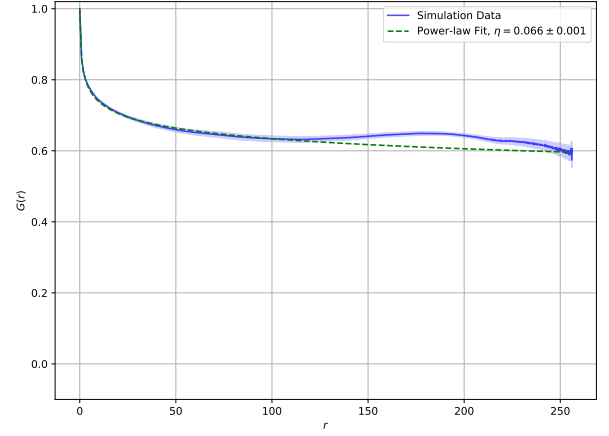
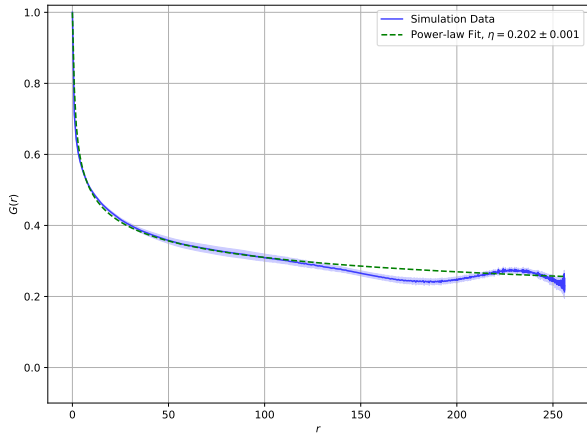
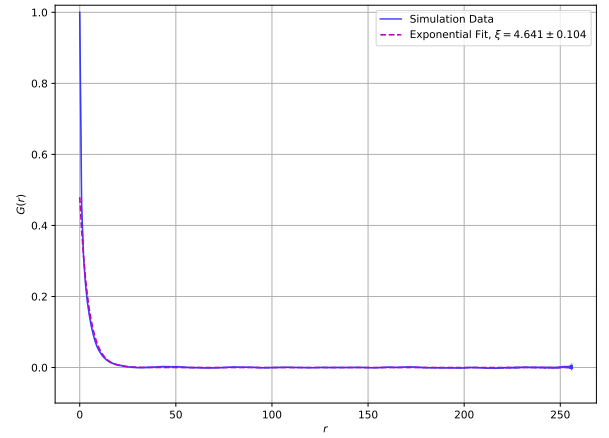
(a) $T = 0.01$ (b) $T = 0.15$ (c) $T = 0.185$ (d) $T = 0.22$

Figure 5: Correlation functions for the polycrystalline model with $\epsilon = 0.0$ and $\gamma = 1.0$ at different temperatures with lattice size $L = 256$.

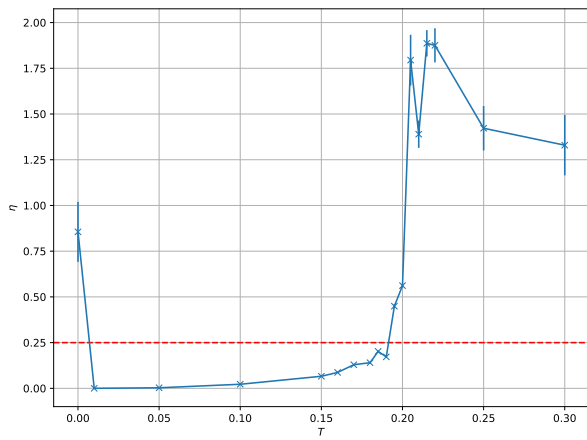
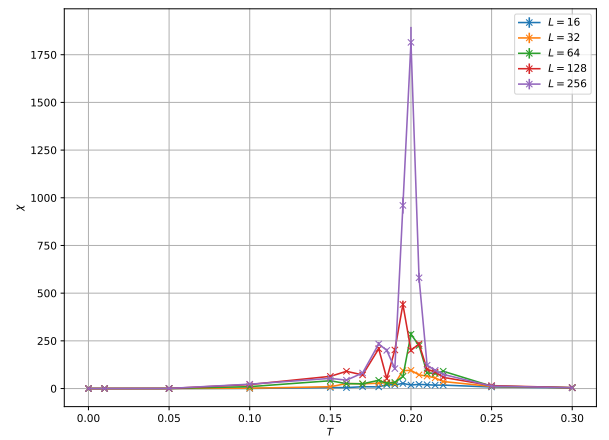
(a) η_6 (b) χ_6

Figure 6: Exponent η_6 and susceptibility χ_6 of the polycrystalline model with $\epsilon = 0.0$ and $\gamma = 1.0$ as a function of the temperature for $L = 256$, respectively, $L = 16, 32, 64, 128, 256$.

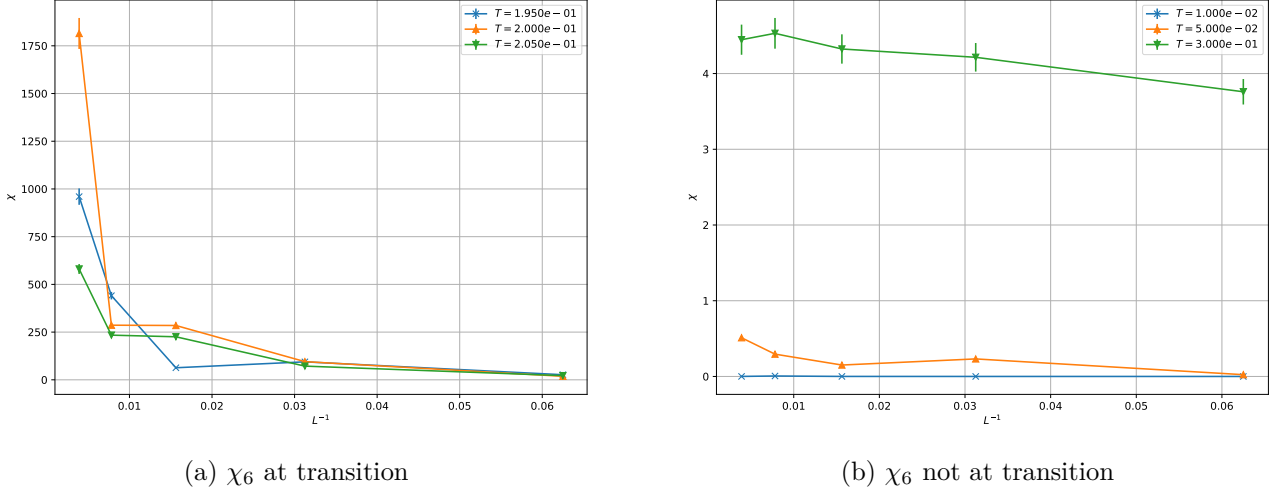


Figure 7: Susceptibility χ_6 as a function of the inverse lattice size L^{-1} for the polycrystalline model with $\epsilon = 0.0$ and $\gamma = 1.0$ at different temperatures near the transition (a) and far from the transition (b).

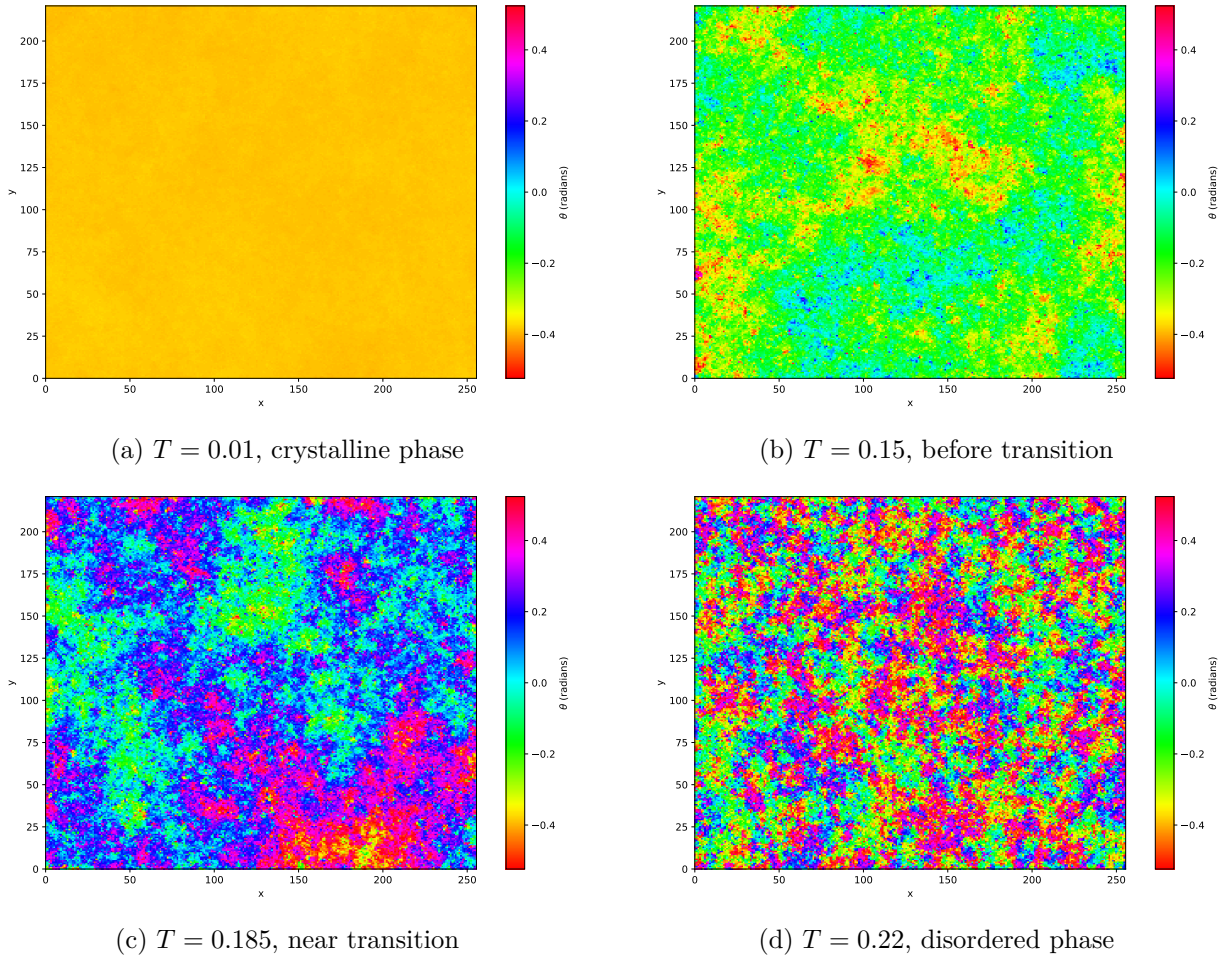


Figure 8: Final lattice configurations for the polycrystalline model with $\epsilon = 0.0$ and $\gamma = 1.0$ at different temperatures with lattice size $L = 256$.

5 Conclusion

In this report, we have explored the orientational ordering phenomena in a coarse-grained model of polycrystalline graphene using Monte Carlo simulations. Our findings indicate the presence of distinct phases characterized by different orientational correlation behaviors consistent with the KTHNY theory of two-dimensional melting.

The validation of our simulation framework against the well-known two-dimensional XY model confirms the accuracy of our numerical methods. The observed BKT transition, marked by a change from algebraic to exponential decay of the correlation function and a corresponding peak in the susceptibility, serves as a benchmark for the next analyses. The study of the polycrystalline graphene model revealed a similar transition scenario when grain-boundary interactions are considered. These results suggest that hexatic orientational order can emerge in two-dimensional polycrystalline systems and could be described within the KTHNY theoretical framework.

Future work could confirm the nature of the transitions observed and refine the estimates of critical temperatures and exponents. Furthermore, integrating random grain extension with different sizes and positions and considering elastic distortions in crystallites could provide a more realistic representation of polycrystalline graphene.

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